

2.6 Numerical Integration: Midpoint, Trapezoid, & Simpson's Rule

Motivation:

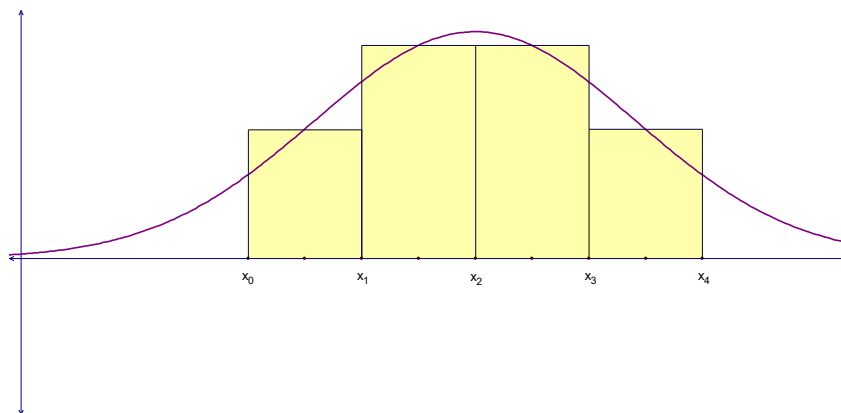
In section 2.5 we learned that some indefinite integrals, like $\int e^{-x^2} dx$, cannot be written in terms of elementary functions. So, a good question to ask would be: How can one calculate $\int_a^b e^{-x^2} dx$ to a high level of accuracy?

Midpoint Rule

In calculus 1 you learned that if we divide $[a, b]$ into n subintervals of length $\Delta x = \frac{b-a}{n}$, then

we have $\int_a^b f(x) dx \approx \sum_{i=1}^n f(x_i^*) \Delta x$ where x_i^* is any point in the i^{th} subinterval which we call $[x_{i-1}, x_i]$ (Note: with this naming scheme we have $x_i = a + i\Delta x$ for $i = 0, 1, \dots, n$).

You also learned that the midpoint rule is the approximation that we obtain when we let $x_i^* = \frac{x_{i-1} + x_i}{2}$. The midpoint rule is pictorially represented below with $n = 4$:



If the curve in purple represents the graph of $f(x)$ then the midpoint rule tells us that the approximate value of $\int_{x_0}^{x_4} f(x) dx$ is given by the sum of the areas of the four yellow rectangles.

The Midpoint Rule: $\int_a^b f(x) dx \approx \Delta x [f(\bar{x}_1) + f(\bar{x}_2) + \dots + f(\bar{x}_n)]$ where $\Delta x = \frac{b-a}{n}$ and $\bar{x}_i = \frac{x_{i-1} + x_i}{2}$.

Example 1: Use the midpoint rule with $n = 2$ to approximate the value of $\int_{-1}^1 e^{-x^2} dx$.

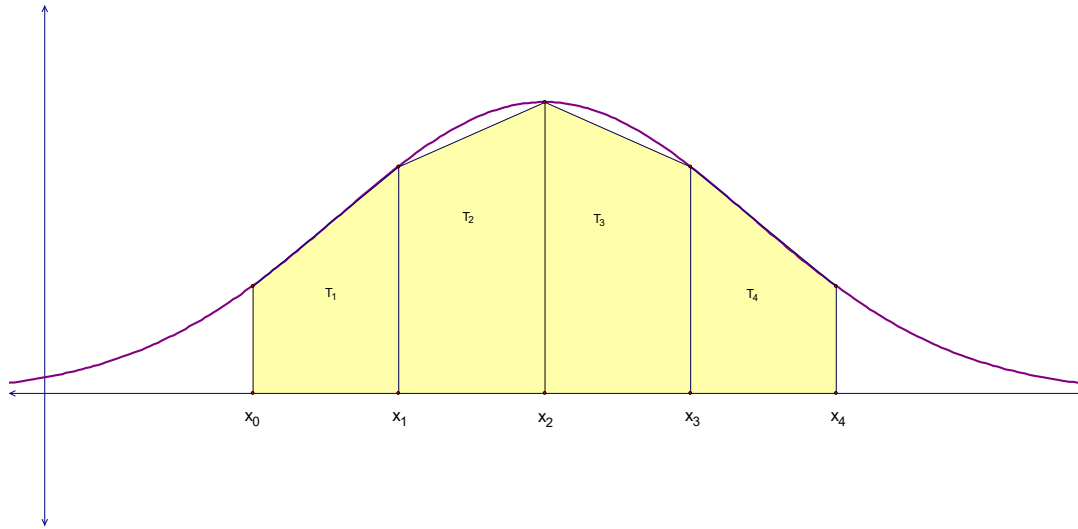
We see that in this case $a = -1$ and $b = 1$. This means that $\Delta x = \frac{1+1}{2} = 1$. This implies that $x_0 = -1$, $x_1 = 0$, and

$x_2 = 1$. So, if we let $f(x) = e^{-x^2}$ by the midpoint rule we see:

$$\int_{-1}^1 e^{-x^2} dx \approx 1 \left(f(\bar{x}_1) + f(\bar{x}_2) \right) = f\left(-\frac{1}{2}\right) + f\left(\frac{1}{2}\right) = e^{-1/4} + e^{-1/4} = \frac{2}{\sqrt[4]{e}} \approx 1.558.$$

Trapezoidal Rule

The idea here is to use trapezoids rather than rectangles to approximate the area. The challenge is to pick the trapezoids so that their areas can be computed quickly and efficiently. The idea (with $n = 4$) of the trapezoidal rule is shown pictorially below:

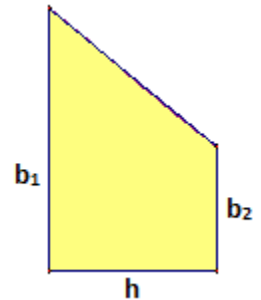


Note: We will use the names Δx and x_i in the same way that we used these names when discussing the midpoint rule.

In the above picture the approximate value of $\int_{x_0}^{x_4} f(x)dx$ is given by the sum of the area of the four trapezoids.

Now, we will compute this sum.

First, recall that the area of a trapezoid is given by the formula: $A = \frac{h}{2}(b_1 + b_2)$ where h is the height of the trapezoid and b_1 and b_2 represent the length of the bases. Using this formula we see that the area of T_1 is $\frac{\Delta x}{2}(f(x_0) + f(x_1))$.



Following this pattern, we see that the sum of the areas of all four trapezoids is:

$\frac{\Delta x}{2}(f(x_0) + f(x_1)) + \frac{\Delta x}{2}(f(x_1) + f(x_2)) + \frac{\Delta x}{2}(f(x_2) + f(x_3)) + \frac{\Delta x}{2}(f(x_3) + f(x_4))$. If we factor out $\frac{\Delta x}{2}$ and combine like terms we see that this expression is equal to:

$\frac{\Delta x}{2}(f(x_0) + 2f(x_1) + 2f(x_2) + 2f(x_3) + f(x_4))$. Generalizing this for arbitrary n leads to the trapezoidal rule.

The Trapezoidal Rule: $\int_a^b f(x)dx \approx \frac{\Delta x}{2}[f(x_0) + 2f(x_1) + 2f(x_2) + \dots + 2f(x_{n-1}) + f(x_n)]$ where $\Delta x = \frac{b-a}{n}$ and $x_i = a + i\Delta x$ for $i = 0, 1, \dots, n$.

Example 2:

Use the trapezoidal rule with $n = 2$ to approximate the value of $\int_{-1}^1 e^{-x^2} dx$.

We see that in this case $a = -1$ and $b = 1$. This means that $\Delta x = \frac{1+1}{2} = 1$. This implies that $x_0 = -1$, $x_1 = 0$, and

$x_2 = 1$. So, if we let $f(x) = e^{-x^2}$ by the trapezoidal rule we see:

$$\int_{-1}^1 e^{-x^2} dx \approx \frac{1}{2} [f(-1) + 2f(0) + f(1)] = \frac{1}{2} [e^{-1} + 2 + e^{-1}] = 1 + e^{-1} \approx 1.368$$

Question to the Class:

Clearly the approximation yielded by the midpoint rule and the trapezoidal rule gets better as the value of n increases. A good question to ask is: “Is there a certain value of n that will guarantee that my approximation is within an error of E from the actual area?”

If we trust that the calculator can find $\int_{-1}^1 e^{-x^2} dx$ to a high level of accuracy (we get $\int_{-1}^1 e^{-x^2} dx \approx 1.494$ from the calculator), we see that the amount the midpoint rule was off (for example 1) was approximately:

$|1.558 - 1.494| = .064$ and the amount the trapezoidal rule (for example 2) was off was approximately:

$|1.368 - 1.494| = .126$. So, in this example the midpoint rule appeared to do a better job.

Theorem: Suppose $|f''(x)| \leq K$ for all x in $[a, b]$. If E_T and E_M are the errors in the Trapezoidal and Midpoint rules, then

$$|E_T| \leq \frac{K(b-a)^3}{12n^2} \quad \text{and} \quad |E_M| \leq \frac{K(b-a)^3}{24n^2}.$$

Proof: The proof of this theorem will be withheld.

Remark: From this theorem we see that we have a better upper bound on the error from the Midpoint rule. So, in this sense the midpoint rule is better. However, there are some examples in which the trapezoidal rule will do just as well as the midpoint rule of approximating the value of a definite integral. It is important to understand that the above theorem is not giving you the error; rather, it is giving you an upper bound on the error (in actuality the error might be much smaller than the upper bound).

Note: Interestingly, applied mathematicians (namely numerical analysts) would view the Midpoint and Trapezoidal rules as roughly the same with respect to how “good” they are since both errors can be bound by a constant times $\frac{1}{n^2}$.

Example 3:

- (a) Suppose we approximate $\int_{-1}^1 e^{-x^2} dx$ with the midpoint rule and $n = 5$. Give an upper bound for the error involved in this approximation.

From the theorem we know we must find K so that $|f''(x)| \leq K$ for every x in $[-1, 1]$ (in this case $f(x) = e^{-x^2}$). We note that $f''(x) = e^{-x^2} (4x^2 - 2)$. Now, using the closed interval method (on page 268) from Calc 1 we can find the global max and global min of $f''(x)$ on $[-1, 1]$. After doing this we see that $|f''(x)| \leq 2$. So, we can take $K = 2$.

Now, from the theorem we see that an upper bound for the error is:

$$|E_M| = \frac{2(1 - (-1))^3}{24(5)^2} = \frac{16}{600} = \frac{2}{75} = 0.02\bar{6}.$$

- (b) How large should we take n if we want to use the midpoint rule and obtain an approximation of $\int_{-1}^1 e^{-x^2} dx$ that is within $\frac{1}{1000}$ of the actual value of $\int_{-1}^1 e^{-x^2} dx$?

The desired n is such that $\frac{16}{24n^2} < \frac{1}{1000}$. Solving this inequality for $n > 0$ we have:

$$\frac{16}{24n^2} < \frac{1}{1000} \Rightarrow 16000 < 24n^2 \Rightarrow \frac{2000}{3} < n^2 \Rightarrow \sqrt{\frac{2000}{3}} < n. \text{ Then since } 26 > \sqrt{\frac{2000}{3}}, \text{ we know}$$

we can take $n \geq 26$ to achieve the desired result.

Simpson's Rule

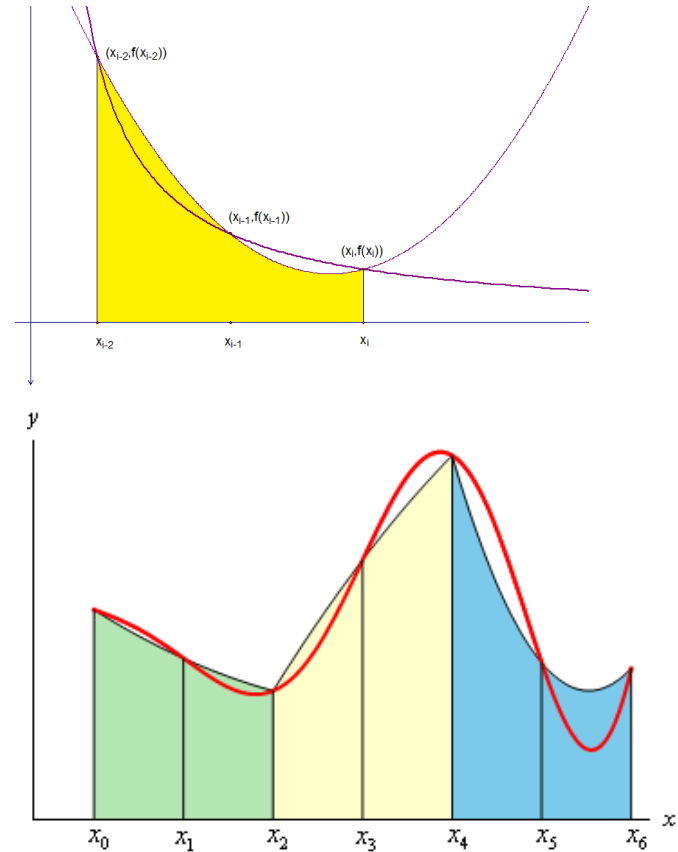
The idea here is to use parabolas instead of straight line segments (we use straight line segments for the trapezoidal rule) to approximate a curve. We will use the names Δx and x_i in the same way that we used these names when discussing the midpoint and trapezoidal rules. Since 3 noncolinear points determine a parabola we use the points $(x_{i-2}, f(x_{i-2}))$, $(x_{i-1}, f(x_{i-1}))$, and $(x_i, f(x_i))$ for $i = 2, 4, 6, \dots, n$ to determine a parabola. Then, to approximate the area under the curve, we find the area bounded between the x -axis and the parabola (from x_{i-2} to x_i). This is illustrated below:

Note: It is important to understand that the area above in yellow is simply one "section" of Simpson's rule. In actuality there can be many points in the partition of the x -axis.

The image below represents the approximation over six subintervals, using 3 parabolas (each different color is an approximation by 1 such parabola). While not every approximation looks like a parabola, they really are quadratics.

Since any parabola that we find in Simpson's rule will exist over two of the subintervals (which $[a, b]$ is divided into) it must be the case that n is even when we apply Simpson's rule.

Due to a lack of lecture time we will not derive Simpson's rule. We now present Simpson's rule and the bound on the error (without proof) when Simpson's rule is applied.



Simpson's Rule: $\int_a^b f(x)dx \approx \frac{\Delta x}{3} [f(x_0) + 4f(x_1) + 2f(x_2) + \dots + 2f(x_{n-2}) + 4f(x_{n-1}) + f(x_n)]$ where n is even and $\Delta x = \frac{b-a}{n}$.

Theorem: Suppose $|f^{(4)}(x)| \leq K$ for all x in $[a, b]$. If E_s is the error involved in using Simpson's rule then:

$$|E_s| \leq \frac{K(b-a)^5}{180n^4}.$$

Remark: One should note that the error bound for Simpson's rule has the potential to be *much* smaller than the error bounds for the Trapezoidal and Midpoint rules.

Example 4:

- (a) Use Simpson's rule with $n = 4$ to approximate the value of $\int_{-1}^1 e^{-x^2} dx$ (round your answer to four decimal places).

We see that in this case $a = -1$ and $b = 1$. This means that $\Delta x = \frac{1+1}{4} = \frac{1}{2}$. This implies that $x_0 = -1$,

$x_1 = -\frac{1}{2}$, $x_2 = 0$, $x_3 = \frac{1}{2}$, and $x_4 = 1$. So, if we let $f(x) = e^{-x^2}$ by Simpson's rule we see:

$$\int_{-1}^1 e^{-x^2} dx \approx \frac{1}{6} [f(x_0) + 4f(x_1) + 2f(x_2) + 4f(x_3) + f(x_4)] \approx 1.4944.$$

- (b) Suppose that you know that if $f(x) = e^{-x^2}$ then $|f^{(4)}(x)| \leq 12$ for any x in $[-1, 1]$. Use this fact to find an upper bound on the error of the approximation you found in (a).

By the theorem the error is bounded above by $|E_s| \leq \frac{12(1-(-1))^5}{180(4)^4} = \frac{384}{46080} = \frac{1}{120} = 0.008\bar{3}$.